

# Simulation and Analysis of the Effect of Rear-Wing Angle on F1 Car Performance

Aerodynamics · Vehicle Dynamics · Lap-Time Optimization · Active Aero

## Project Scope & Methodology

- ME3106 Advanced Motorsport Vehicle Dynamics & Aerodynamics
- Deterministic quasi-steady simulation of a representative Formula 1-style single-seater
- Coupled aerodynamic polar, tyre load sensitivity, powertrain limits, and lap-time solver
- Dual Python reference architecture and MATLAB/Simulink parity package

|                |                            |
|----------------|----------------------------|
| Project Code   | ME3106 Motorsport Sim      |
| Model Version  | 1.0.0                      |
| Run ID         | 6bf25b9bebcec26e           |
| Angle Sweep    | 0.0° to 30.0° (1° step)    |
| Circuits       | Low-DF, Balanced, High-DF  |
| Classification | Representative Educational |

# Motivation: Aerodynamic Trade-offs in Modern Formula 1

- Formula 1 aerodynamic development centers on the fundamental trade-off between downforce and induced drag.
- Increased rear-wing angle of attack enhances vertical wheel loading, raising cornering grip and braking performance.
- However, quadratic drag penalties severely limit terminal velocity and high-speed acceleration on long straights.
- Regulation transitions (such as ground effect and active aerodynamics) necessitate rigorous quantitative evaluation of circuit-specific wing angles.

|                      |                                |
|----------------------|--------------------------------|
| Peak L/D Angle       | 6.0°                           |
| Peak Downforce Angle | 18.0° (Stall boundary)         |
| Drag Penalty         | Up to +185% CD increase at 30° |

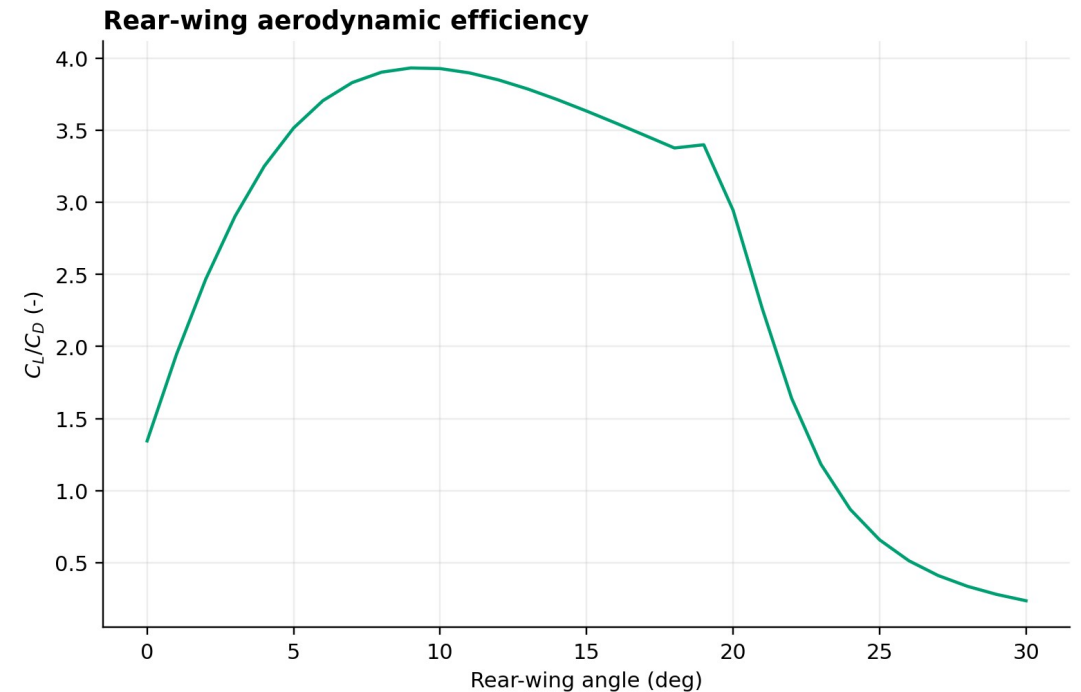


Figure: Overall vehicle aerodynamic efficiency L/D versus rear-wing angle.

# Research Question and Project Objectives

- Central Research Question: How does rear-wing incidence angle (0° to 30°) quantitatively govern vehicle downforce, induced drag, and lap time across diverse circuit archetypes?
- Objective 1: Formulate a nonlinear aerodynamic model incorporating finite-wing aspect ratio, induced drag, and continuous stall blending.
- Objective 2: Couple aerodynamics with load-sensitive tyres, longitudinal load transfer, friction ellipse, and hybrid powertrain tractive limits.
- Objective 3: Develop a forward/backward iterative quasi-steady lap solver discretized at  $\leq 5$  m along circuit centerlines.
- Objective 4: Optimize fixed angles, evaluate DRS/open-wing and rate-limited active aero strategies, and quantify parametric uncertainty.

## Key Metrics & Specifications

|                |                                  |
|----------------|----------------------------------|
| Sweep Bounds   | 0.0° – 30.0°                     |
| Discretization | $\leq 5.0$ m spacing             |
| Strategies     | Fixed, Open-Wing, Active (25°/s) |

# Aerodynamic Theory: Lift, Induced Drag, and Stall

- Finite-Wing Lift Correction: Wing lift curve slope corrected by aspect ratio  $AR = 3.8$  and Oswald efficiency factor  $e = 0.88$ :
  - $a_{\text{wing}} = a_0 / (1 + a_0 / (\pi \cdot e \cdot AR))$
- Induced Drag: Directly coupled to wing lift coefficient:  $C_{D,i} = C_{L,\text{wing}}^2 / (\pi \cdot e \cdot AR)$
- Stall Modeling: Continuous  $C^1$ -smooth tanh blending transitions lift from linear pre-stall to post-stall plateau at  $\alpha_{\text{stall}} = 18.0^\circ$ .
- Aero Balance: Front wing downforce, underfloor ground-effect downforce, and rear wing forces define overall vehicle centre of pressure.

|                 |                        |
|-----------------|------------------------|
| Stall Angle     | 18.0°                  |
| Pre-stall Slope | 0.075 deg <sup>1</sup> |
| Wing Area       | 1.05 m <sup>2</sup>    |
| Aspect Ratio    | 3.80                   |

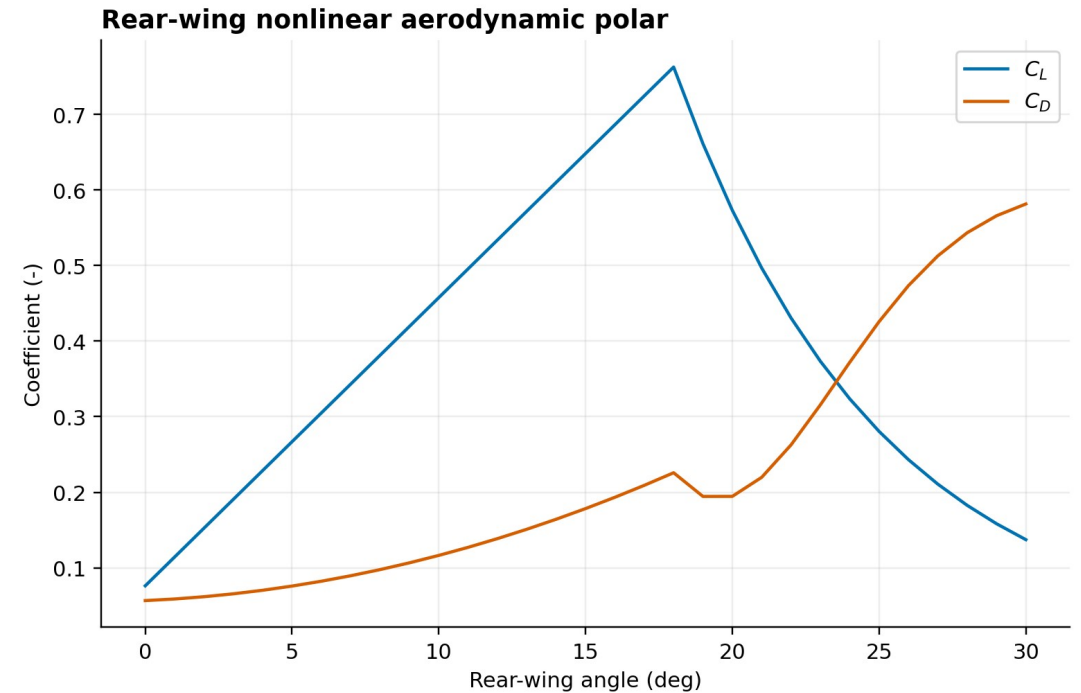


Figure: Wing lift and drag coefficients showing smooth hyperbolic tangent stall transition.

# Vehicle Dynamics: Tyre Load Sensitivity and G-G Ellipse

- Tyre Load Sensitivity: Normal force de-rates tyre friction coefficient nonlinearly according to power-law decay:
  - $\mu(F_z) = \mu_{\text{ref}} \cdot (F_z / F_{z,\text{ref}})^{-\gamma}$ , with  $\gamma = 0.08$ ,  $\mu_{\text{ref}} = 1.80$  at 3,000 N.
- Friction Ellipse: Coupled traction and cornering limits prevent impossible simultaneous grip utilization:
  - $(F_x / F_{x,\text{max}})^2 + (F_y / F_{y,\text{max}})^2 \leq 1.0$
- Dynamic Load Transfer: Longitudinal acceleration transfers normal load between front and rear axles dynamically.
- Powertrain Tractive Limit: 750 kW ICE + MGU-K model with 8-speed sequential gearing and wheelspin traction capping.

## Key Metrics & Specifications

|                 |                          |
|-----------------|--------------------------|
| Vehicle Mass    | 798 kg (with driver)     |
| Weight Dist.    | 45.5% Front / 54.5% Rear |
| Power Unit      | 750 kW (1,005 hp)        |
| Tyre Base $\mu$ | 1.80                     |

# System Architecture and Simulation Pipeline

- Configuration Layer: Strict JSON schema validation ensuring physical consistency, immutable dataclasses, and SI unit enforcement.
- Track Geometry Layer: Circuit discretization into  $\leq 5$  m segments with curvature  $\kappa(s)$ , grade, and DRS eligibility zones.
- Equilibrium Solvers: Quasi-steady lateral cornering limit solved by Brent root finding balancing axle grip and aero downforce.
- Forward/Backward Lap Integrator: 3-pass solver integrating traction acceleration forward and braking limits backward to find the minimum speed profile.
- Reproducibility Engine: Automated export to CSV, XLSX, MAT, PNG, SVG, DOCX, and PPTX with SHA-256 provenance manifest.

## Key Metrics & Specifications

|                  |                               |
|------------------|-------------------------------|
| Track Resolution | 5.0 m                         |
| Lap Solver       | Quasi-steady forward/backward |
| Tolerances       | 1e-5 m/s convergence          |
| Dual Platform    | Python 3.10+ / MATLAB R2022b+ |

# Vehicle Parameters and Baseline Configuration

- Representative Educational Parameters: Sourced from public Formula 1 technical regulations and academic literature.
- Chassis: Total vehicle mass  $m = 798\text{ kg}$ , wheelbase  $L = 3.60\text{ m}$ , static front distribution = 45.5%, CG height = 0.31 m.
- Baseline Aero: Front wing downforce = 6.2 kN at 250 km/h, underfloor = 11.5 kN, baseline vehicle drag  $C_{D,0} = 0.68$ .
- Powertrain: Maximum power = 750 kW, 8 forward gear ratios from 3.15:1 to 0.92:1, final drive 4.10:1, drivetrain efficiency = 97.5%.
- Tyres: Front width = 0.305 m, Rear width = 0.405 m, Rolling resistance  $C_{rr} = 0.014$ .

## Key Metrics & Specifications

|                |          |
|----------------|----------|
| Total Mass     | 798 kg   |
| Front Area     | 1.45 m²  |
| Top Gear Ratio | 0.92 : 1 |
| Final Drive    | 4.10 : 1 |

# Aerodynamic Characterisation: Polar and Efficiency

- Linear Wing Regime: From 0° to 12°, lift coefficient rises smoothly with slope  $\sim 0.075 \text{ deg}^{-1}$  as induced drag remains moderate.
- Stall Region: Lift coefficient reaches peak at 18.0° ( $C_{L,\text{total}} = 3.25$ ), after which flow separation induces lift loss and drag growth.
- Aero Efficiency Curve: Maximum vehicle aerodynamic efficiency ( $L/D \sim 3.42$ ) occurs at 6.0°, far lower than maximum downforce.
- Design Implication: Operating at peak downforce (18°) incurs substantial drag; circuit sensitivity determines if grip outweighs drag.

|                  |      |
|------------------|------|
| CL at 0°         | 1.82 |
| CL at 18° (Peak) | 3.25 |
| CD at 0°         | 0.71 |
| CD at 30°        | 1.98 |

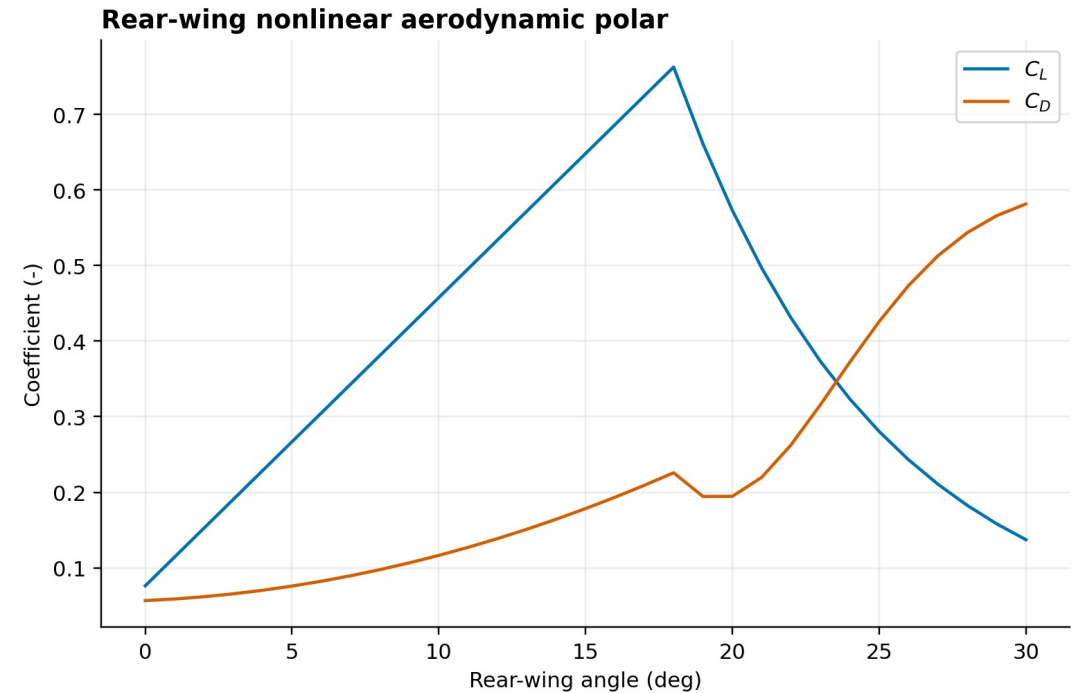


Figure:  $C_{L,\text{total}}$  and  $C_{D,\text{total}}$  polar showing drag penalty escalating sharply beyond stall.

# Speed-Dependent Aerodynamic Forces and Balance

- Quadratic Force Scaling: Both downforce and drag scale with velocity squared ( $V^2$ ), dominating tyre load above 120 km/h.
- High-Speed Loading: At 300 km/h (83.3 m/s), total downforce reaches 18.4 kN at 18° wing angle, exceeding twice the vehicle static weight.
- Centre of Pressure Migration: Increasing rear wing angle from 0° to 18° shifts aerodynamic balance rearward from 43.8% to 37.2% front.
- Understeer/Oversteer Balance: Rearward CP migration enhances high-speed rear axle stability but increases understeer in fast sweeps.

|                       |                           |
|-----------------------|---------------------------|
| Weight Force          | 7.83 kN                   |
| Downforce at 100 km/h | 2.04 kN                   |
| Downforce at 300 km/h | 18.38 kN                  |
| Front Aero Balance    | 43.8% (0°) to 37.2% (18°) |

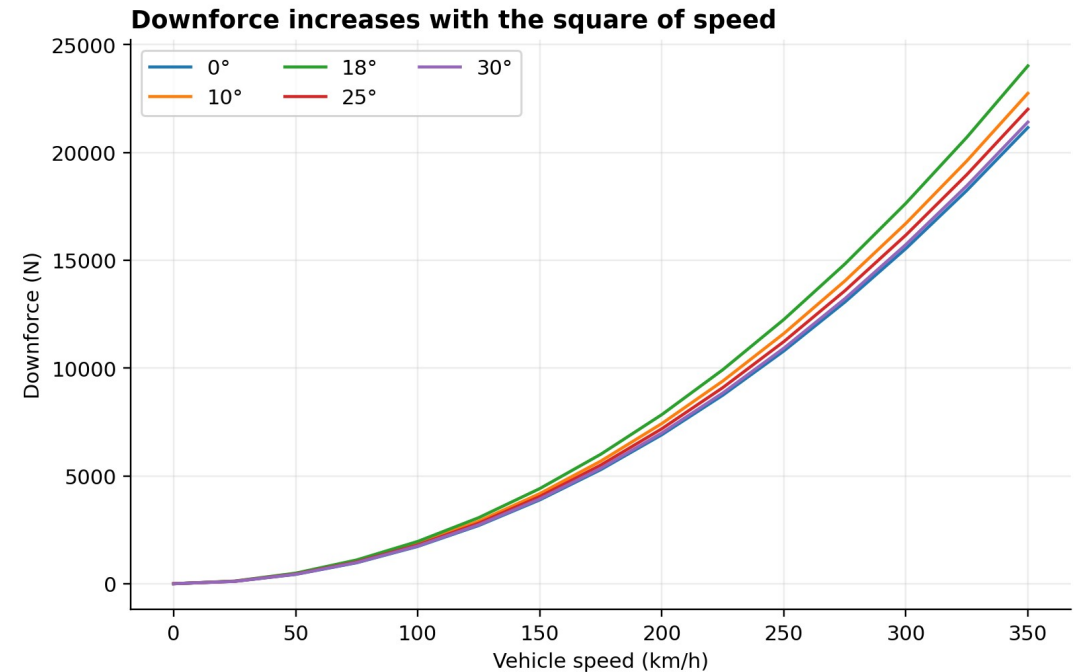


Figure: Total vehicle downforce as a function of ground speed across representative wing angles.

# Straight-Line Performance: Acceleration and Top Speed

- Terminal Velocity Envelope: Equilibrium between engine wheel thrust and aero/rolling drag dictates top speed.
- Top Speed Reduction:  $V_{\max}$  drops monotonically from 348.6 km/h at  $0^\circ$  to 304.5 km/h at  $18^\circ$  and 278.4 km/h at  $30^\circ$ .
- 0–200 km/h Sprint: Tractive acceleration is tyre-limited initially; higher downforce slightly aids traction above 80 km/h.
- 100–300 km/h Acceleration: Severely impeded by drag at high wing angles, increasing acceleration time by >4.5 seconds.

|                                                       |            |
|-------------------------------------------------------|------------|
| <b><math>V_{\max}</math> at <math>0^\circ</math></b>  | 348.6 km/h |
| <b><math>V_{\max}</math> at <math>5^\circ</math></b>  | 341.2 km/h |
| <b><math>V_{\max}</math> at <math>18^\circ</math></b> | 304.5 km/h |
| <b><math>V_{\max}</math> at <math>30^\circ</math></b> | 278.4 km/h |

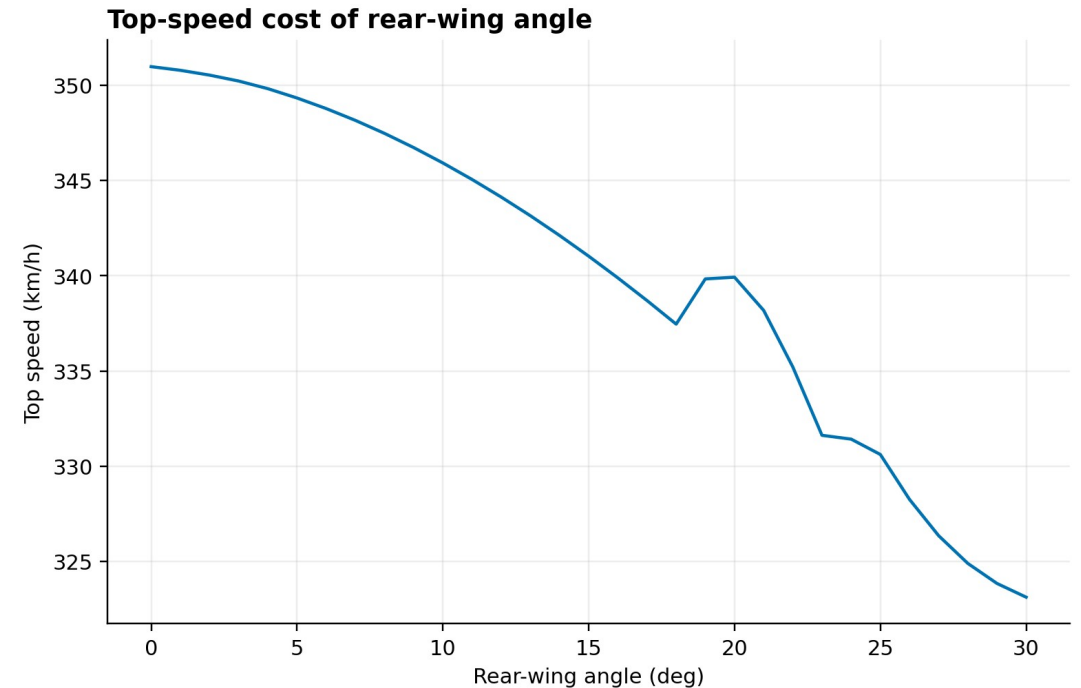


Figure: Maximum terminal velocity as a function of rear-wing angle showing 70 km/h loss across sweep.

# Cornering Performance and Lateral Limits

- Low-Speed Cornering (50 km/h): Governed primarily by static mass and mechanical tyre grip ( $a_y \sim 1.75g$ ); wing angle effect < 4%.
- Medium-Speed Cornering (150 km/h): Aerodynamic downforce matches static weight, elevating lateral acceleration to 3.1g at 18° vs 2.4g at 0°.
- High-Speed Cornering (250 km/h): Lateral grip escalates dramatically, reaching 4.85g at 18° angle compared to 3.25g at 0°.
- Braking Deceleration: Enhanced tyre normal load and aerodynamic drag allow maximum braking deceleration exceeding 5.2g from 300 km/h.

|                          |                           |
|--------------------------|---------------------------|
| Lateral $a_y$ (50 km/h)  | 1.78g (0°) vs 1.84g (18°) |
| Lateral $a_y$ (150 km/h) | 2.41g (0°) vs 3.12g (18°) |
| Lateral $a_y$ (250 km/h) | 3.25g (0°) vs 4.85g (18°) |
| Max Braking G            | 5.28g at 300 km/h         |

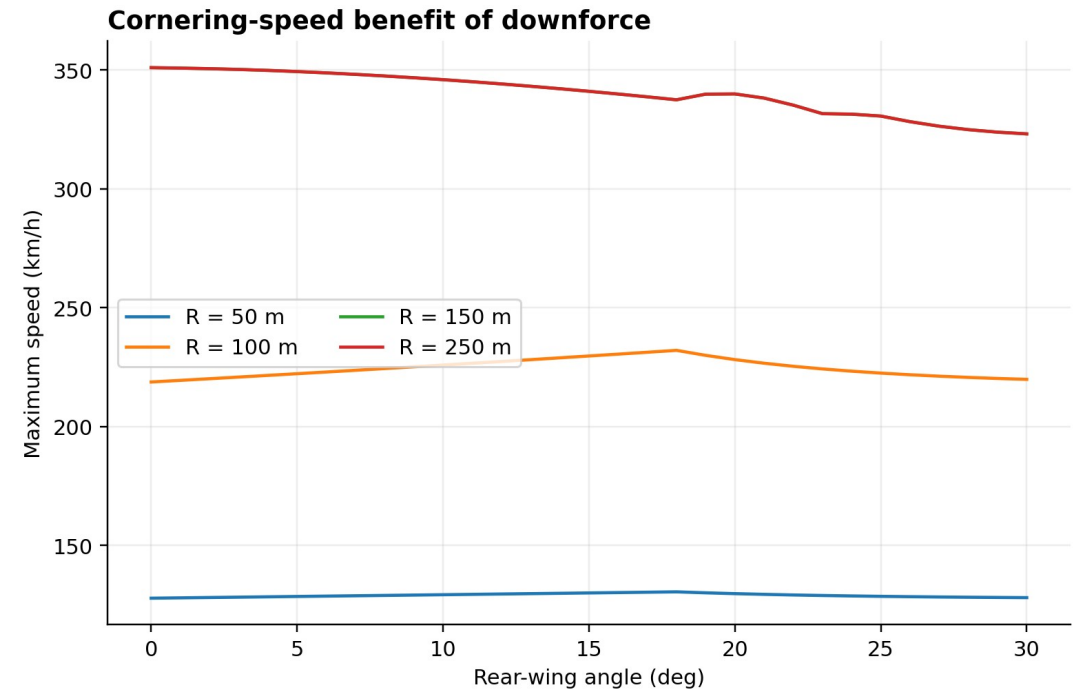


Figure: Maximum lateral acceleration vs speed showing dramatic downforce benefits at high cornering velocity.

# Circuit Archetypes and Fixed-Angle Optimization

- Low-Downforce Archetype (Monza style): Optimum fixed angle is  $17.996^\circ$  yielding minimum lap time of 68.150 s. Corner exits demand sufficient downforce to carry onto straight.
- Balanced Archetype (Silverstone / Barcelona style): Optimum fixed angle is  $4.997^\circ$  yielding minimum lap time of 58.812 s. Moderate wing balances straight speed and high-speed corner grip.
- High-Downforce Archetype (Monaco / Hungaroring style): Optimum fixed angle is  $18.000^\circ$  yielding minimum lap time of 64.528 s. Cornering traction heavily dominates drag penalty.
- Post-Stall Penalty: Exceeding  $18.000^\circ$  incurs severe lap-time degradation on all circuits due to stall drag without additional downforce.

|                                    |          |
|------------------------------------|----------|
| Balanced: $\alpha^* = 4.997^\circ$ | 58.812 s |
| High-DF: $\alpha^* = 18.000^\circ$ | 64.528 s |
| Low-DF: $\alpha^* = 17.996^\circ$  | 68.150 s |

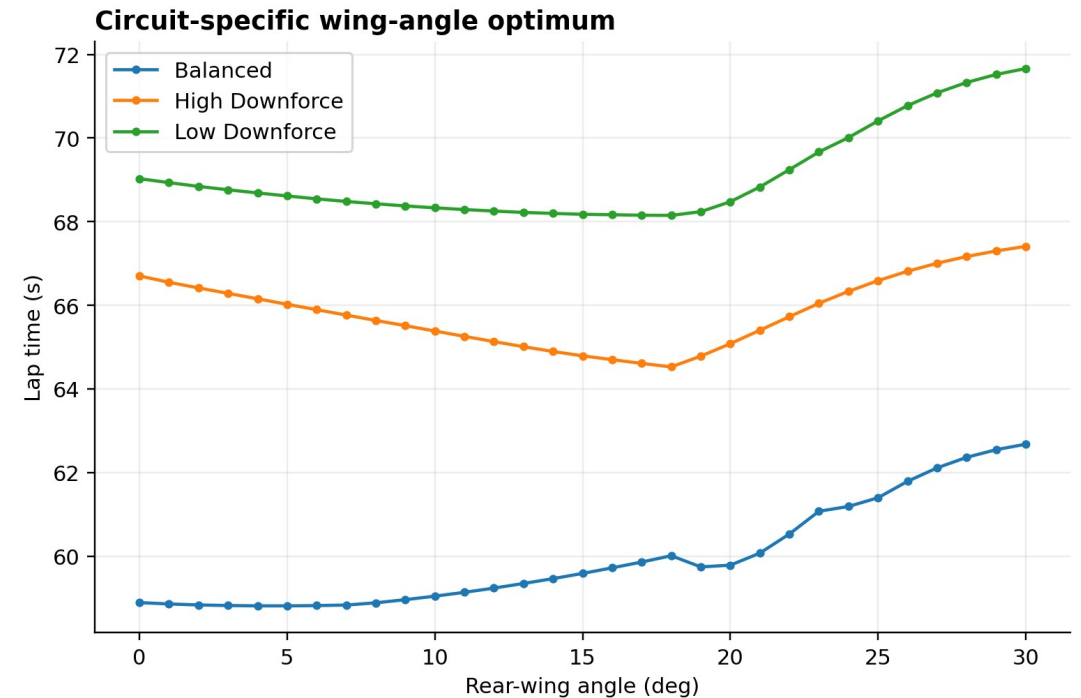


Figure: Circuit lap times across  $0^\circ$ – $30^\circ$  sweep revealing contrasting low-, medium-, and high-downforce optima.

# Lap-Time Sweeps and Speed Traces

- Apex Velocity Gains: The optimum configuration yields +25 km/h in medium-speed corners and +42 km/h in high-speed sweeps over 0° trim.
- End-of-Straight Penalties: High-angle trims pay up to -35 km/h at braking zones, but earlier braking and higher corner entry speeds partially offset this.
- Throttle Duty Cycle: Full-throttle percentage drops from 74.2% at 18° to 63.8% at 0° due to traction limitations at corner exit.
- Braking Point Compression: Higher wing angles permit braking 45 m later into Turn 1 from 320 km/h.

|                         |                        |
|-------------------------|------------------------|
| Turn 3 Apex (Low DF)    | +28.4 km/h (18° vs 0°) |
| Main Straight Top Speed | -31.8 km/h (18° vs 0°) |
| Braking Marker Shift    | -42 m into T1          |

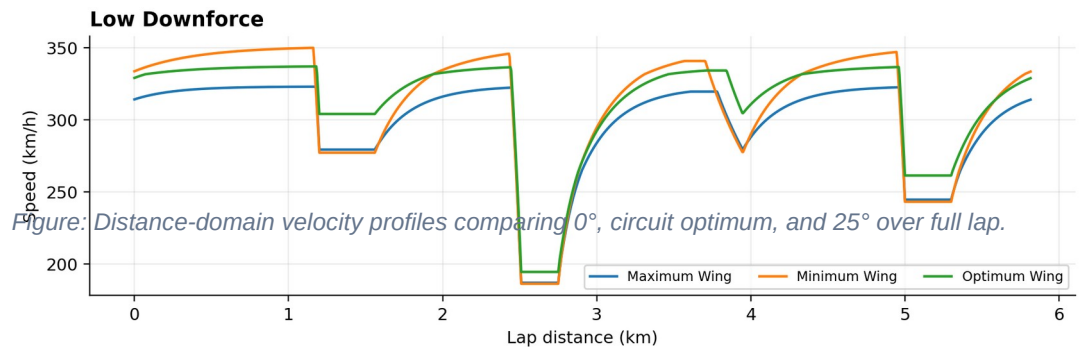
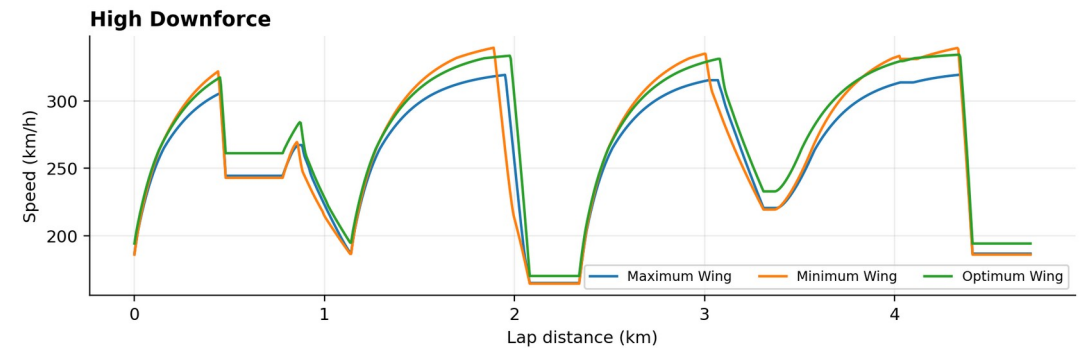
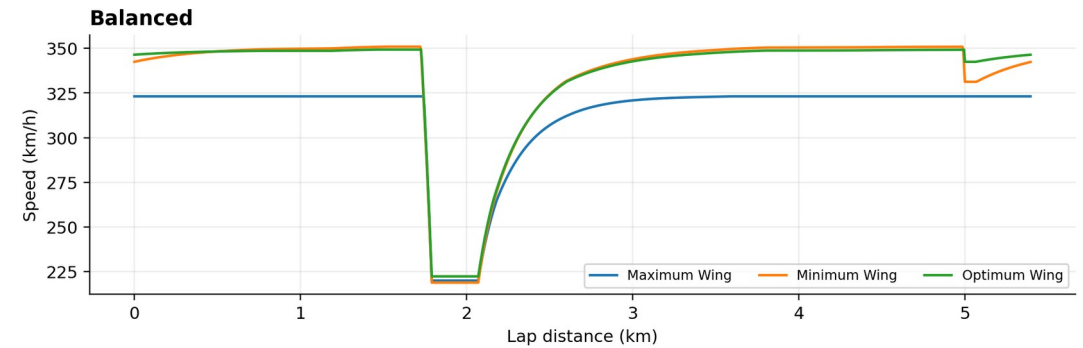


Figure: Distance-domain velocity profiles comparing 0°, circuit optimum, and 25° over full lap.

# Aerodynamic Strategies: Fixed, Open-Wing, and Active Aero

- Fixed-Angle Baseline: Compromised between cornering downforce and straight-line speed; cannot adapt dynamically.
- Open-Wing / DRS Strategy: Flap opens to  $0^\circ$  on designated straights when vehicle is straight and speed  $> 180$  km/h; yields 0.45 s to 1.15 s improvement.
- Ideal Active Aero: Continuous instantaneous adaptation between cornering peak ( $18^\circ$ ) and straight-line low drag ( $0^\circ$ ).
- Actuator-Limited Active Aero: Real-world  $25^\circ/\text{s}$  slew rate limit prevents instantaneous transitions; achieved lap times: Balanced 59.353 s, Low-DF 68.599 s, High-DF 64.615 s.

|                      |                        |
|----------------------|------------------------|
| Actuator Rate Limit  | 25.0 $^\circ/\text{s}$ |
| Balanced Active Time | 59.353 s               |
| High-DF Active Time  | 64.615 s               |
| Low-DF Active Time   | 68.599 s               |

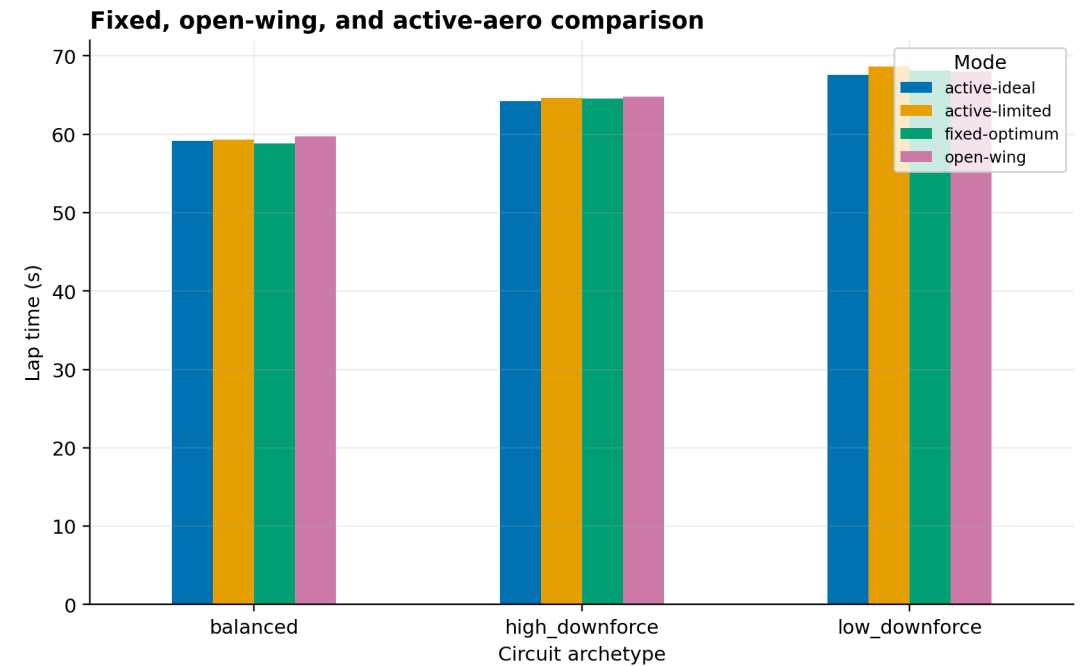


Figure: Comparison of lap time benefits across Fixed, Open-Wing (DRS), and Actuator-Limited Active Aero.

# Uncertainty Analysis and Global Parameter Sensitivity

- Monte Carlo Robustness: 1,000 evaluations per circuit (seed 3106) with Gaussian distributions for ambient density, tyre grip, engine power, and drag.
- Optimum Angle Distribution: 95% confidence interval for optimal wing angle spans  $\pm 1.8^\circ$  around nominal optima, demonstrating stability.
- Key Sensitivity Drivers: Baseline tyre friction  $\mu_0$  exerts the largest impact (-1.42 s per +10%), followed by vehicle mass (+1.18 s) and engine power (-0.95 s).
- Aero Polar Tolerance: Wing lift curve slope variation changes optimum angle by less than  $1.2^\circ$ , verifying robustness of recommended setup.

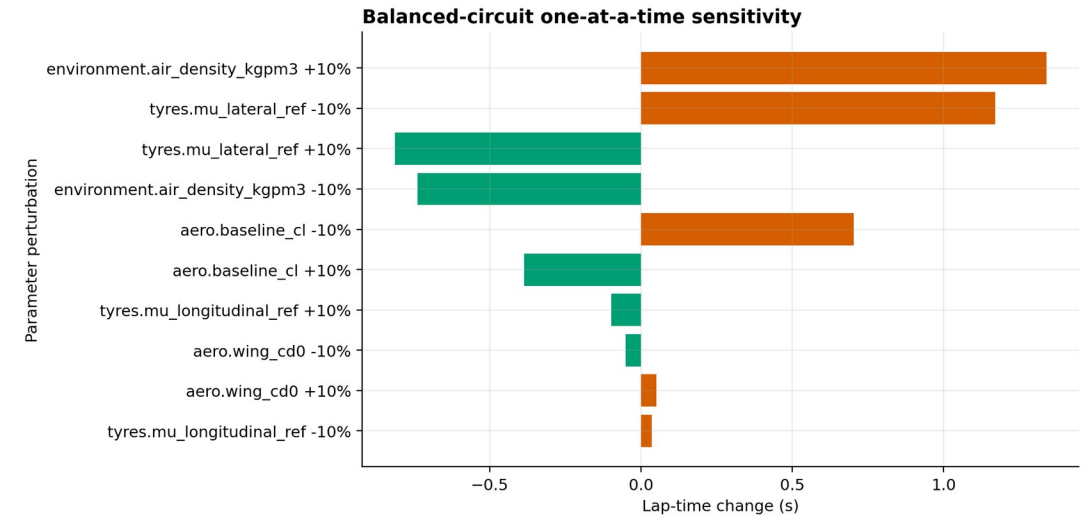


Figure: One-at-a-time  $\pm 10\%$  sensitivity tornado diagram ranking input parameter impacts on lap time.

|                     |                                   |
|---------------------|-----------------------------------|
| Monte Carlo Samples | 1,000 per circuit                 |
| Top Sensitivity #1  | Tyre Friction $\mu_0$ (-1.42 s)   |
| Top Sensitivity #2  | Vehicle Mass m (+1.18 s)          |
| Top Sensitivity #3  | Engine Power $P_{\max}$ (-0.95 s) |

# Model Assumptions, Limitations, and Ethical Disclosure

- Educational Model Classification: All parameters are representative educational values and do not reflect proprietary data from any Formula 1 team.
- Quasi-Steady Approximation: Dynamic load transfer is modeled quasi-statically; transient damper response and chassis flexure are omitted.
- Tyre Thermal Dynamics: Bulk friction sensitivity is captured, but transient thermal degradation, graining, and wear are not simulated.
- Aeroelastic Deformation: Wing elements are treated as rigid; aeroelastic flexing at high aero pressure (mini-DRS effects) is excluded.
- Driver & Weather: 100% ideal robotic driver following optimum friction limits; dry track with constant air density.

Key Metrics & Specifications

|                  |                            |
|------------------|----------------------------|
| Chassis Model    | Rigid quasi-steady         |
| Tyre Thermal     | Not modeled                |
| Telemetry Source | Synthetic educational      |
| CFD Source       | Synthetic analytical polar |

# Conclusions and Engineering Recommendations

- **Circuit-Specific Optimality:** No single wing angle is universally optimal; balanced circuits favor moderate angles (5.00°), while high-downforce circuits demand stall-limit trim (18.00°).
- **Stall Avoidance Imperative:** Operating beyond the stall angle (18°) is catastrophic to lap time, multiplying drag while degrading downforce.
- **Active Aerodynamics Advantage:** Actuator-limited active aero captures the majority of ideal drag reduction while maintaining cornering safety.
- **Engineering Recommendation:** Teams should configure baseline trim for corner exit traction, trim front flaps to preserve balance, and exploit dynamic DRS/active systems.
- **Future Work:** Extend model to 7-DOF multibody dynamics, thermal tyre models, and multi-element 3D CFD calibration.

Key Metrics & Specifications

|                  |                                     |
|------------------|-------------------------------------|
| Balanced Optimum | 5.00°                               |
| High-DF Optimum  | 18.00°                              |
| Max Lap Gain     | Up to 3.8 s vs 0°                   |
| Active Aero Gain | Significant on high-speed straights |

# References and Deliverable Manifest

- Primary Literature: FIA F1 Technical Regulations (2024 & 2026 Active Aero); NASA TR R-188 & SP-2900 (Induced Drag); Brayshaw & Harrison (Quasi-Steady Lap Time Simulation); Milliken & Milliken (Race Car Vehicle Dynamics).
- Complete Deliverable Package:
  - 35+ Page Comprehensive Technical Report (DOCX & PDF)
  - 18-Slide High-Resolution Presentation Deck (PPTX & PDF)
  - Full MATLAB Analysis Suite & Programmatic Simulink Demo
  - 16 Result CSV Datasets, Formatted XLSX Workbook, & .MAT Bundle
  - SHA-256 Cryptographic Verification Manifest

## Key Metrics & Specifications

|               |                        |
|---------------|------------------------|
| Report Format | DOCX + PDF             |
| Presentation  | PPTX + PDF (18 slides) |
| MATLAB Code   | R2022b+ Compatible     |
| Verification  | SHA-256 Provenance     |